

Research Statement

CIFRE PhD — CREST, Institut de Mathématiques de Toulouse, Generali France

*“When Diversification Fails”: **changer le titre pour mettre plus accent sur les maths***

*Regime-Conditional Factor Dependence, Supervised Capital Proxy,
and Deep Hedging for Insurance Total Portfolio Approach*

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1. INDUSTRIAL CONTEXT AND MOTIVATION

1.1. Starting point: 2022, a dependence crisis

The year 2022 was a major stress test for insurance asset managers. Unlike previous crises — which were essentially isolated shocks on a single asset class — 2022 saw equities and bonds fall simultaneously, inter-asset correlations turn positive, and traditional asset-class diversification break down entirely. For a life insurer such as Generali France, this episode exposed a fundamental structural flaw: Strategic Asset Allocation (SAA) models assume stable, linear dependence structures between economic factors, whereas the empirical evidence shows that these dependencies reverse abruptly in stress regimes.

This observation directly motivates the object of this thesis: *how **to model** and to manage an insurance balance sheet when factor dependence becomes unstable, non-linear, and regime-dependent?*

1.2. The current SAA process and its limitations

How the SAA process works today. The current SAA process at Generali France operates in three sequential steps, each carrying structural limitations.

Step 1 — Economic Scenario Generator (ESG). A proprietary ESG produces thousands of economic trajectories (interest rates, credit spreads, equity, real estate, inflation) over a long horizon, calibrated on historical data with fixed correlations averaged across all regimes. These trajectories feed all downstream actuarial calculations.

Step 2 — SCR calculation via nested LSMC. The Solvency Capital Requirement (SCR) is the regulatory capital under Solvency II, defined as the 99.5% one-year Value-at-Risk of net asset value change. Its calculation requires evaluating the Best Estimate of Liabilities (BEL) — the risk-neutral present value of all future policyholder cash flows — under each of the 10,000 real-world scenarios produced by the ESG. Since each BEL evaluation requires a nested risk-neutral simulation (typically 1,000 inner paths), the raw computational cost is $10,000 \times 1,000 = 10$ million simulations. The industry-standard solution — Least-Squares Monte Carlo (LSMC, Krah et al., 2018) — replaces the inner simulation by a polynomial proxy $\hat{V}^L(\mathbf{x}) = \sum_k \beta_k \phi_k(\mathbf{x})$ fitted offline on a design-of-experiment. Even so, a full SCR calculation takes several hours, making dynamic optimisation loops computationally infeasible.

Step 3 — Annual Markowitz optimisation. The SCR is treated as a fixed constraint in a mean-variance optimisation over asset classes, revised once per year.

Four structural limitations.

1. **Regime-blind dependence.** The ESG calibrates correlations on long historical averages. In 2022, the equity/bond correlation switched from -0.3 (diversifying) to $+0.6$ (concentration risk) within weeks. A model with fixed correlations cannot anticipate such reversals: the Markowitz optimum for a low-inflation regime is a high-risk portfolio in a stagflation regime.

2. **LSMC proxy limitations.** The polynomial LSMC proxy is accurate only within the region covered by the design-of-experiment, which is constructed without regime conditioning. Precision degrades precisely in the stress zones most relevant for capital decisions.
3. **Static, siloed decision-making.** Markowitz optimises a single-period mean-variance criterion without anticipating future regime transitions or internalising hedging costs. Allocation and hedging are decided separately, yielding a suboptimal joint policy for an insurer whose critical losses are convex.
4. **Private asset valuation.** Private equity, private debt, infrastructure and real estate constitute a growing share of Generali's portfolio. Their quarterly, smoothed NAV valuations create an illusion of low correlation with public markets: in 2008, private equity NAVs remained stable for 6–12 months after the crash before correcting sharply. This artificial smoothing understates true economic dependence in stress regimes and introduces a lag bias in the SCR calculation. Furthermore, the irrevocable capital commitments of closed-end funds are incompatible with dynamic rebalancing frameworks that treat all assets as freely tradeable.

1.3. The TPA as target framework: what is missing: **a enlever je pense, ou alors a diminuer tres fortement**

The **Total Portfolio Approach (TPA)** is the management framework that Generali France aims to adopt to progressively replace the current asset-class SAA at local level. Rather than optimising over asset classes, TPA evaluates each decision through its marginal contribution to total balance sheet risk via **common economic factors**: Duration, Inflation, Growth, Liquidity, Carry. Empirical studies on sovereign wealth funds that have already adopted TPA (Future Fund, CPP Investments) report approximately +1% excess return per year relative to classical SAA (Thinking Ahead Institute, 2025; CAIA, 2024).

However, TPA currently rests on simplistic dependence assumptions: linear, Gaussian, non-regime-conditional. This thesis provides the missing quantitative foundations to make TPA operational for an insurer under Solvency II. The **regime-conditional factor dependence** is the transversal thread: it is first modelled and simulated (Axis 1), then translated into fast capital costs via a supervised proxy (Axis 2), and finally integrated into a joint allocation-hedging policy (Axis 3).

Extension to liability-side and joint factors. The TPA framework as currently applied — including in the sovereign wealth fund implementations serving as reference (Future Fund, CPP Investments) — operates exclusively on **asset-side factors** (Duration, Inflation, Growth, Liquidity, Carry). For an insurance company, this is structurally insufficient: the balance sheet is jointly driven by factors that exclusively affect liabilities (surrender rate dynamics, longevity, policyholder behaviour) and by **cross-cutting factors** that simultaneously affect both the asset and liability sides. The most prominent example is the **yield curve**: a parallel shift simultaneously reprices the bond portfolio *and* the Best Estimate of Liabilities, creating complex convexity interactions that cannot be captured by an asset-only factor decomposition. This thesis therefore extends the TPA factor space to include these joint factors, providing a genuinely *balance-sheet-wide* factor framework.

A what-if toolbox for allocation decisions. A key operational consequence of modelling factor trajectories stochastically (Axis 1) is the ability to explore the full spectrum of extreme scenarios systematically — adverse shocks (stress regimes: stagflation, credit crunch, equity crash) as well as scenarios more favourable than expected (yield normalisation, above-trend growth). This produces an **allocation reaction function**: a structured mapping from any projected factor scenario — whether originating from the asset side, the liability side, or joint cross-cutting factors such as the yield curve — to the optimal portfolio response. The result is a *what-if plan* computed ex ante, enabling the investment team to anticipate all market configurations rather than reacting to them ex post. This reaction function is a direct output of the Axis 3 joint policy optimisation and constitutes one of the principal industrial deliverables of the thesis.

2. RESEARCH QUESTIONS

In the spirit of the Total Portfolio Approach (Thinking Ahead Institute, 2025; CAIA, 2024), the objective of the thesis is to propose a global framework as well as quantitative foundations for decision making in the context of complex and high dependencies among risks, such as common economic factors: Duration, Inflation, Growth, Liquidity, Carry, among others. More precisely, a regime-conditional factor dependence will be developed and studied along three interconnected axis: it is first modeled and simulated using generative methodologies (Axis 1), then translated into capital costs via computational methods and supervised proxy (Axis 2), and finally integrated into a joint allocation-hedging policy (Axis 3). These 3 axis are summarized below.

Research Question 1 (Regime-conditional factor dependence: from GAN, VAE and SGM to formal guarantees). *How to simulate the joint distribution of different risk factors conditionally on the current macroeconomic regime — identified in real time by a sequential filter — while capturing the non-linear, asymmetric tail dependencies relevant for extreme risk? Different methodologies will be tested : Generative Adversarial Network (GAN), Variational autoencoders (VAE) and Score-Based Generative Model (SGM), and analyzed/compared depending on different criteria : interpolation, explicability, interpretability. Specifically: (i) how to formally validate the conditional dependence structure learned by a regime-conditional GAN, VAE or SGM, using copula specification tests; and (ii) how to obtain W_2 -convergence guarantees when upgrading to a Score-Based Generative Model (SGM) with sequentially estimated regime states?*

Research Question 2 (Supervised SCR computation, and proxy for dynamic decision-making). *How to compute the insurance SCR and to construct a proxy expressed as a function of the risk factors, the current regime, and key liability states? First, an analysis of the dependence of the risks in the tail of the distribution will be conveyed (beyond classical gaussian distribution) in order to study the impact of model risk on the calculation of extreme quantiles and to ensure a sufficient accuracy in decision-relevant zones (near the regulatory threshold, under stress regimes). In passing, the relevance of calculating a 99.5% quantile (although mandated by the regulator) could be investigated and discussed. A focus on the computation of a proxy will be done, with controlled error bounds per regime, in order to be able to it into a dynamic optimisation loop.*

Research Question 3 (Joint allocation and deep hedging policy). *How to learn a dynamic optimal decision policy jointly optimising risk factor exposures and explicit hedging positions (deep hedging), under capital and liquidity constraints including private asset engagement limits, in the presence of unstable, regime-conditional factor dependencies, so as to reduce the tail risk of the insurance balance sheet? Different optimisation criteria could be investigated beyond the classical Markowitz mean-variance criteria: minimiser une mesure de risque directement, ou maximiser une fonction objectif (utilité d'un capital terminal) mais sous contraintes de ne pas dépasser un seuil de risque donnée par une mesure de risque avec fonction de distorsion, ou par TailVaR ou autre. On peut y intégrer une variante avec Reinforcement learning pour prendre en compte incertitude du model.*

3. RELATED WORK AND POSITIONING

A REGARDER la littérature recente et les ref qui y sont associees. NOTAMMENT: -j'ai ajoute les papiers dans dropbox

pour generative model de time series avec des sauts , regarder le papier recent (j'ai ajoute dans dropbox)
Schrodinger bridges with jumps for time series generation

Stefano De Marco, Huyen Pham, Davide Zanni

<https://arxiv.org/abs/2602.20011>

je mets ici l'abstract car ca me semble bien dans notre topic: "We study generative modeling for time series using entropic optimal transport and the Schrodinger bridge (SB) framework, with a focus on applications in finance and energy modeling. Extending the diffusion-based approach of Hamdouche, Henry-Labordere, Pham, 2023, we introduce a jump-diffusion Schrodinger bridge model that allows for discontinuities in the generative dynamics. Starting from a Schrodinger bridge entropy minimization problem, we reformulate the task as a stochastic control problem whose solution characterizes the optimal controlled jump-diffusion

process. When sampled on a fixed time grid, this process generates synthetic time series matching the joint distributions of the observed data. The model is fully data-driven, as both the drift and the jump intensity are learned directly from the data. We propose practical algorithms for training, sampling, and hyperparameter calibration. Numerical experiments on simulated and real datasets, including financial and energy time series, show that incorporating jumps substantially improves the realism of the generated data, in particular by capturing abrupt movements, heavy tails, and regime changes that diffusion-only models fail to reproduce. Comparisons with state-of-the-art generative models highlight the benefits and limitations of the proposed approach."

pour le calcul de SCR proxy numerique avec rapidité de calcul: cf les travaux de these de Florian Bourgey (cadre credit) (j'ai ajoute la these dans dropbox)

cf les points suivants de la these de Florian "nous examinons plusieurs methodes d'approximations stochastiques à la fois pour le calcul de mesures de risques financiers. Comme les formules explicites sont rarement disponibles pour de telles quantités, le besoin d'approximations analytiques rapides, efficaces et fiables est d'une importance capitale pour les institutions financières. Nous visons ainsi à donner un large aperçu de ces méthodes d'approximation et nous nous concentrons sur trois approches distinctes. Dans la première partie, nous étudions plusieurs méthodes d'approximation Monte Carlo multi-niveaux et les appliquons à deux problèmes pratiques : l'estimation de quantités impliquant des espérances imbriquées (comme la marge initiale). nous analysons les propriétés d'optimalité asymptotique des estimateurs multi-niveaux correspondants et démontrons numériquement leur supériorité par rapport à une méthode de Monte Carlo classique. Dans la deuxième partie, motivés par les nombreux exemples issus de la modélisation en risque de crédit, nous proposons un cadre général de méta-modélisation pour de grandes sommes de variables aléatoires de Bernoulli pondérées, qui sont conditionnellement indépendantes par rapport à un facteur commun X . Notre approche générique est basée sur la décomposition en polynômes du chaos du facteur commun et sur une approximation gaussienne. Les estimations d'erreur L2 sont données lorsque le facteur X est associé à des polynômes orthogonaux classiques."

pour optimisation : cf travaux Emmanuela Rosazza : Shortfall Systemic Risk Measures, Capital Allocation Rules and Generalized Collapse to the Mean: Theory and Practice (cf papiers dans dropbox).

cf également le papier d'Anthony, CRRA utility maximization under dynamic risk constraints, 2013. "The problem of optimal investment with CRRA (constant, relative risk aversion) preferences, subject to dynamic risk constraints on trading strategies, is the main focus of this paper. Several works in the literature, which deal either with optimal trading under static risk constraints or with VaR-based dynamic risk constraints, are extended. The market model considered is continuous in time and incomplete, and the prices of financial assets are modeled by Ito processes. The dynamic risk constraints, which are time and state dependent, are generated by a general class of risk measures. Optimal trading strategies are characterized by a quadratic BSDE. Within the class of time consistent distortion risk measures, a three-fund separation result is established. Numerical results emphasize the effects of imposing risk constraints on trading."

3.1. Generative models and regime-conditional dependence

Ne pas se limiter au GAN - presenter aussi alternatives VAE-SGM. Presenter des suggestions de "mesures" pour comparer les methodologies

Generative Adversarial Networks applied to financial series have demonstrated their capacity to capture multivariate stylized facts: heavy tails, volatility clustering, and asymmetries (Wiese et al., 2020, QuantGAN; Yoon et al., 2019, TimeGAN). Conditional GAN architectures (cGAN) allow generation to be conditioned on external variables — here, the macroeconomic regime. However, GANs suffer from *mode collapse* and provide no formal guarantees on the quality of the generated distribution, limiting their regulatory auditability.

Copula theory (Sklar, 1959) provides a complementary framework for validating learned dependence. By Sklar's theorem:

$$P(\mathbf{F}_t \leq \mathbf{f}) = C(F_1(f_1), \dots, F_d(f_d)),$$

where C is the copula capturing the dependence structure independently of the marginals. Fermanian (2013, 2022) has developed specification tests for conditional copulas, providing the statistical tools to formally validate that a GAN has correctly captured the conditional tail dependence coefficient λ_k per regime.

Score-Based Generative Models (SGMs) offer a generative alternative with formal convergence guarantees. Strasman, Ocello et al. (TMLR, 2024) establish explicit bounds on the Wasserstein-2 distance as a function of the noise schedule $g(t)$:

$$\text{KL}(\hat{p} \| p^*) \leq \int_0^T g(t)^2 \varepsilon_{\text{score}}(t) dt + \varepsilon_{\text{init}}. \quad (1)$$

Gentiloni-Silveri and Ocello (ICML, 2025) extend these guarantees to non-log-concave distributions — notably Gaussian mixtures corresponding to distinct market regimes.

Identified gap. Neither conditional GANs nor conditional SGMs have been applied to simulating the joint distribution of TPA factors with real-time regime identification. For the GAN: the absence of formal conditional tail dependence validation. For the SGM: the absence of W_2 bounds for the conditional case when the regime state is estimated sequentially with inference error. Both contributions constitute Axis 1.

3.2. Insurance liability proxy models

The LSMC framework of Krah et al. (2018) is the industry reference for life insurance proxy models, approximating the BEL by polynomial regression on a pre-computed design-of-experiment. This approach presents two critical limitations: it does not condition the SCR on the current macroeconomic regime, and its accuracy degrades precisely in the stress zones most relevant for the allocation decision. Neural network approaches (Huber & Schmock, 2022) have demonstrated that a well-specified proxy can achieve 1–3% accuracy with inference times below 10ms.

Identified gap. Existing proxies are constructed independently of the regime and TPA factors, and none accounts for the NAV-lag bias of private assets in the SCR calculation. A regime-conditioned, factorially coherent proxy with adaptive sample weighting and controlled error bounds does not exist in the literature. This is the contribution of RQ2.

3.3. Deep RL, dynamic allocation and deep hedging

Wekwete (2023) demonstrated the applicability of Deep RL (PPO, SAC) to insurance ALM in stochastic environments. Buehler et al. (2019) introduced *deep hedging*: learning an optimal hedging strategy via a neural network minimising a convex risk measure, without market completeness assumptions. This approach has been extended to extreme risk hedging (Carbonneau, 2021; Cao et al., 2023) and multi-asset portfolios under stress regimes.

Identified gap. Learning a joint allocation-and-hedging policy — fed by an unstable dependence model and a fast SCR proxy, respecting private asset engagement constraints, within a TPA insurance framework under Solvency II — does not exist in the literature. This is the contribution of RQ3.

4. METHODOLOGY AND DELIVERABLES

4.1. Axis 1 — Regime-conditional factor dependence (Years 1–2)

Axis 1 proceeds in two methodological stages corresponding to a progressive increase in theoretical rigour.

Stage 1 — Regime-conditional GAN (Year 1). Let $z_t \in \{1, \dots, K\}$ be the macroeconomic regime identified in real time by a particle filter (SMC, Chopin & Papaspiliopoulos, 2020). A conditional GAN is trained to simulate the joint distribution of TPA factors conditionally on the current regime:

$$\mathbf{F}_{t+1} | z_t = k \sim G_\theta(\varepsilon, k), \quad \varepsilon \sim \mathcal{N}(0, I), \quad (2)$$

where G_θ is the generator and the discriminator D_ϕ distinguishes real from synthetic scenarios regime by regime. The advantage of the GAN is total flexibility: no parametric assumption on the dependence structure is imposed, allowing the model to learn asymmetries and non-linearities in tail dependence under stress. The conditional tail dependence coefficient

$$\lambda_k = \lim_{u \rightarrow 1} P(F_i > F_{i,k}^{-1}(u) | F_j > F_{j,k}^{-1}(u), z_t = k)$$

is estimated empirically and validated via Fermanian’s (2013) specification tests for conditional copulas.

Private asset treatment. The factor vector includes a *vintage lag factor* $\tilde{F}_t^{\text{PA}} = \text{NAV}_{t-1}^{\text{PA}}$ capturing the quarterly smoothing bias of private equity, real estate and infrastructure NAVs. The GAN learns the true economic dependence between private and public assets, including the delayed correction dynamic observed in 2008 and 2020.

Stage 2 — SGM with W_2 guarantees (Year 2). The GAN, while empirically powerful, suffers from mode collapse and provides no formal guarantees — limiting regulatory auditability. The second stage replaces or complements the GAN with a conditional SGM, building on the theoretical framework of Strasman, Ocello et al. (TMLR, 2024) and Gentiloni-Silveri & Ocello (ICML, 2025). The conditional OU forward process for regime $z_t = k$ is:

$$d\mathbf{x}^\tau = -\mathbf{x}^\tau d\tau + \sqrt{2}g_k(\tau)dW_\tau, \quad \tau \in [0, T], \quad (3)$$

with regime-specific noise schedule $g_k(\tau)$. The convergence bounds give:

$$W_2(\hat{p}_k, p_k^*)^2 \leq C \int_0^T g_k(\tau)^2 \varepsilon_{\text{score}}(\tau) d\tau,$$

even for non-log-concave distributions corresponding to distinct market regimes.

Theoretical contribution. (i) Empirical validation of tail dependence λ_k generated by the GAN via Fermanian (2013) specification tests. (ii) Extension of the W_2 bounds of Gentiloni-Silveri & Ocello (ICML 2025) to the conditional SGM case with online HMM — the key open question: how do these bounds evolve when regime z_t is estimated sequentially with particle filter inference error?

Python deliverable — factor_engine module.

- HMMRegimeFilter: online SMC filter, sequential update of $P(z_t | \mathbf{F}_{1:t})$ on real-time Bloomberg data.
- RegimeConditionedGAN: conditional GAN trained per regime; tail dependence validation via copula specification tests (Fermanian, 2013).
- RegimeConditionedSGM: conditional SGM with explicit W_2 bounds; regime-specific noise schedule $g_k(\tau)$.
- FactorScenarioGenerator: generation of N regime-conditional factor scenarios via GAN (stage 1) or SGM (stage 2); backtesting on 2008, 2020 and 2022 crises.

4.2. Axis 2 — Supervised SCR proxy (Year 2)

Model. Let $\mathbf{P}_t \in \mathbb{R}^p$ be the liability state vector: effective duration, average guaranteed rate, observed lapse ratio, and the private asset vintage lag $\tilde{F}_{t-1}^{\text{PA}}$ capturing NAV smoothing bias. We learn a supervised proxy:

$$\widehat{\text{SCR}}(t) = f_\theta(\mathbf{F}_t, z_t, \mathbf{P}_t), \quad (4)$$

where f_θ is a neural network (MLP + attention) trained on a design-of-experiment $\{(\mathbf{F}^{(i)}, z^{(i)}, \mathbf{P}^{(i)}), \text{SCR}^{(i)}\}$ computed offline via Prophet/MoSés. The key methodological innovation is **adaptive weighting**: scenarios close to the regulatory threshold and in stress regimes receive higher training weight, concentrating accuracy where the decision is most sensitive. The generalisation bound satisfies:

$$|\text{SCR}(t) - f_\theta(\mathbf{F}_t, z_t, \mathbf{P}_t)| \leq \varepsilon_k(n_k, \delta), \quad (5)$$

where $\varepsilon_k(n_k, \delta)$ decreases with the number of training points n_k in regime k and with confidence level $1 - \delta$.

Contribution. First SCR proxy jointly conditioned on the macroeconomic regime, TPA factors, and private asset NAV lag, with empirical error bounds validated across portfolio ranges including Fonds Euro and Euro-Croissance. Inference in < 5 ms versus several hours for the full Prophet/MoSés calculation.

Python deliverable — scr_proxy module.

- LiabilityDataPipeline: Prophet/MoSés interface for automated actuarial design-of-experiment generation.

- **SCRProxyModel**: MLP + attention network trained on 5,000+ Generali scenarios with adaptive regime weighting.
- **SCRCalculator**: near-instantaneous SCR computation (< 5 ms) for Market, Interest Rate and Equity modules.
- **Validation report**: relative error and confidence intervals per regime; stress tests on 2022 and novel scenarios.

4.3. Axis 3 — Joint allocation and deep hedging policy (Year 3)

Model. The agent state is $s_t = (\mathbf{F}_t, \hat{C}_{z_t}, z_t, \mathbf{P}_t, \widehat{\text{SCR}}_t)$, where $\hat{C}_{z_t}$ is the regime-estimated copula from Axis 1 and $\widehat{\text{SCR}}_t$ the proxy from Axis 2. The action is the pair $(\mathbf{w}_t, \mathbf{h}_t)$, where $\mathbf{w}_t \in \Delta^{d-1}$ is the TPA factor allocation vector and $\mathbf{h}_t \in \mathcal{H}$ is the hedging position (puts, overlays, convexity structures). The policy π maximises:

$$\max_{\pi} \mathbb{E}_{\pi} \left[\sum_{t=0}^T \left(r(\mathbf{w}_t, \mathbf{h}_t, \mathbf{F}_{t+1}) - \lambda_1 \max(0, \widehat{\text{SCR}}_t - \bar{C}) - \lambda_2 c(\mathbf{h}_t) - \lambda_3 \|\mathbf{w}_t - \mathbf{w}_{t-1}\|_1 - \lambda_4 \mathbf{1}[\Delta w_t^{\text{PA}} > 0] \right) \right], \quad (6)$$

where r is the risk-adjusted return, $\bar{C}$ the regulatory capital budget, $c(\mathbf{h}_t)$ the hedging cost (option premium, bid-ask spread), and the term $\lambda_4 \mathbf{1}[\Delta w_t^{\text{PA}} > 0]$ penalises new private asset commitments to reflect their irrevocable, illiquid nature.

Integrated deep hedging. Unlike classical deep hedging (Buehler et al., 2019) which targets payoff replication, the hedging component here aims to reduce the $\text{CVaR}_{0.99}$ of the capital deficit while internalising hedging costs. The regime-conditional factor dependence of Axis 1 is directly embedded in the simulation environment, forcing the policy to be robust to dependence regime switches — including the private asset delayed correction dynamic.

Python deliverable — dl_allocator module.

- **TPA_Environment**: Gym environment coupling `factor_engine` and `scr_proxy`; simulates regime-switching trajectories including private asset NAV dynamics.
- **AllocationHedgingAgent**: PPO/SAC agent with dual head (factor allocation + hedging); trained on 10+ years of simulated multi-regime data.
- **RebalancingTrigger**: automated detection of trigger events (regime change, SCR threshold breach, target drift $> \Delta$); GenAMBridge interface for automated order execution.
- **StrategyBacktester**: full backtesting on 2008, 2020, 2022; RoRC, regime-conditional Sharpe, SCR CVaR, hedging cost metrics.

5. SYSTEM ARCHITECTURE

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All modules are packaged in the Python library `solvency-tpa`, deployed in production at Generali with Sphinx documentation and pytest unit tests.

Module 1	Module 2	Module 3
factor_engine	scr_proxy	dl_allocator
HMMRegimeFilter (SMC)	LiabilityDataPipeline	TPA_Environment (Gym)
RegimeConditionedGAN	SCRProxyModel	AllocationHedging
RegimeConditionedSGM	(MLP + attention)	Agent (PPO/SAC)
FactorScenarioGen.	SCRCalculator	RebalancingTrigger
<i>Input:</i> Bloomberg rates, spreads, vol, private NAV lags	<i>Input:</i> Prophet/MoSes + M1 factors + private NAV lag	<i>Input:</i> M1 + M2 outputs
<i>Output:</i> regime-conditional scenarios + copulas $\hat{C}_k$	<i>Output:</i> SCR in < 5 ms + error bounds per regime	<i>Output:</i> allocation + hedge + GenAM orders

Tech stack: Python 3.11+, PyTorch (neural proxy and RL agent), pyvinecopulib / scipy (copulas), particles (SMC, Chopin), gymnasium / stable-baselines3 (RL), Bloomberg API. GPU cluster for training; CPU for production inference.

6. EXPECTED CONTRIBUTIONS

Type	Description
<i>Theoretical</i>	(i) Non-asymptotic error bounds on estimated conditional tail dependence λ_k for regime-conditional copulas and GANs (Fermanian); (ii) Extension of W_2 bounds of Gentiloni-Silveri & Ocello (ICML 2025) to the conditional SGM case with online SMC regime estimation (Ocello); (iii) Generalisation bounds for the regime-conditioned SCR proxy with adaptive weighting and private asset NAV lag.
<i>Methodological</i>	(i) HMM-SMC architecture coupled to a regime-conditional GAN/SGM with private asset vintage lag factor (module 1); (ii) Supervised SCR proxy conditioned on regime and TPA factors with adaptive weighting (module 2); (iii) Dual allocation-and-hedging agent under private asset engagement constraints (module 3).
<i>Industrial</i>	Python library solvency-tpa in production at Generali: auditable regime-conditional scenario generation, SCR in < 5 ms, TPA allocation with integrated hedging and GenAM interface.
<i>Regulatory</i>	Models with controlled, auditable guarantees — W_2 bounds for the generative module, empirical error bounds for the SCR proxy — directly usable by the ACPR and the Generali Group Risk Division.

7. TIMELINE (36 MONTHS)

Phase	Period	Key deliverables
1	1st year	Bloomberg factor database (20 years) including private asset NAV lags; online HMM-SMC calibration in production; regime-conditional GAN with copula validation on historical crises; <code>factor_engine</code> v1.0; paper submission (Journal of Multivariate Analysis or Finance & Stochastics).
2	2nd year	Actuarial design-of-experiment (Prophet/MoSes); training and validation of regime-conditioned SCR proxy with private asset NAV lag; empirical error bounds per regime and portfolio range; <code>scr_proxy</code> v1.0; paper submission (Insurance: Mathematics and Economics).
3	3rd year	Training of dual allocation-and-hedging agent with private asset engagement constraints; full three-module coupling; multi-regime backtesting (2008, 2020, 2022); <code>dl_allocator</code> v1.0; GenAM interface in production; PhD defence and publications (Quantitative Finance or JFQA).

REFERENCES

- [1] Buehler, H., Gonon, L., Teichmann, J., & Wood, B. (2019). Deep hedging. *Quantitative Finance*, 19(8), 1271–1291.
- [2] CAIA Association. (2024). *Innovation Unleashed: The Rise of Total Portfolio Approach*.
- [3] Cao, J., Chen, J., Hull, J., & Poulos, Z. (2023). Deep hedging of derivatives using reinforcement learning. *The Journal of Financial Data Science*, 5(1), 10–27.
- [4] Carbonneau, A. (2021). Deep hedging of long-term financial derivatives. *Insurance: Mathematics and Economics*, 99, 327–340.
- [5] Chopin, N., & Papaspiliopoulos, O. (2020). *An Introduction to Sequential Monte Carlo*. Springer.
- [6] Embrechts, P., McNeil, A.J., & Straumann, D. (2002). Correlation and dependence in risk management: properties and pitfalls. In *Risk Management: Value at Risk and Beyond*, Cambridge University Press, 176–223.
- [7] Fermanian, J.-D. (2013). An overview of the goodness-of-fit test problem for copulas. In *Copulae in Mathematical and Quantitative Finance*, Springer, 61–89.
- [8] Fermanian, J.-D. (2022). Conditional copulas, dependence modeling and the role of conditioning. *Annals of Statistics*.
- [9] Goodfellow, I. et al. (2014). Generative adversarial networks. *Advances in Neural Information Processing Systems*, 27.
- [10] Hamilton, J.D. (1989). A new approach to the economic analysis of nonstationary time series. *Econometrica*, 57(2), 357–384.
- [11] Huber, M., & Schmock, B. (2022). A neural network approach to efficient valuation of large portfolios of variable annuities. *Insurance: Mathematics and Economics*, 105, 96–116.
- [12] Krah, A.-S., Nikolic, Z., & Korn, R. (2018). A Least-Squares Monte Carlo framework in proxy modeling of life insurance companies. *Risks*, 6(2), 62.
- [13] Nystrup, P., Madsen, H., & Lindstrom, E. (2017). Dynamic portfolio optimization across hidden market regimes. *Quantitative Finance*, 18(1), 83–95.
- [14] Strasman, S., Ocello, A., Boyer, C., Le Corff, S., & Lemaire, V. (2024). An analysis of the noise schedule for score-based generative models. *Transactions on Machine Learning Research (TMLR)*.
- [15] Gentiloni-Silveri, M., & Ocello, A. (2025). Beyond log-concavity and score regularity: Improved convergence bounds for score-based generative models in W_2 -distance. *Forty-Second International Conference on Machine Learning (ICML)*.

- [16] Patton, A.J. (2006). Modelling asymmetric exchange rate dependence. *International Economic Review*, 47(2), 527–556.
- [17] Pham, H. (2009). *Continuous-Time Stochastic Control and Optimization with Financial Applications*. Springer.
- [18] Sklar, A. (1959). Fonctions de repartition a n dimensions et leurs marges. *Publications de l'Institut de Statistique de Paris*, 8, 229–231.
- [19] Thinking Ahead Institute. (2025). *TPA Report: From Vision to Execution*. WTW / CAIA.
- [20] Wekwete, T. (2023). Application of Deep Reinforcement Learning in Asset Liability Management. *Actuarial Society of South Africa Convention Paper*.
- [21] Wiese, M., Knobloch, R., Korn, R., & Kretschmer, P. (2020). Quant GANs: Deep generation of financial time series. *Quantitative Finance*, 20(9), 1419–1440.
- [22] Yoon, J., Jarrett, D., & van der Schaar, M. (2019). Time-series generative adversarial networks. *Advances in Neural Information Processing Systems*, 32.